PYTHON MINI PROJECT

30-Minute Student Task Report

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Task	Two-Player Tic-Tac-Toe Console Game

1. Task Overview

The task is to build a two-player Tic-Tac-Toe game that runs in the console. Two players take turns placing their mark ('X' or 'O') on a 3x3 grid using position numbers 1-9. The program checks after each move whether a player has won, or whether the board is full resulting in a draw. The target flow is DISPLAY BOARD → INPUT MOVE → UPDATE BOARD → CHECK WINNER → SWITCH TURN → REPEAT.

2. Features Implemented

- Represents the board as a list of 9 elements, displayed as a 3x3 grid.
- Uses a function to print the board neatly with row separators.
- Uses a while loop to alternate turns between Player X and Player O.
- Validates that the chosen position is on the board and not already taken.
- Uses a function with all 8 winning combinations to check for a win.
- Detects a draw when the board is full and no winner is found.
- Switches the current player automatically after every valid move.
- Announces the winner by name/symbol or declares a draw at the end.

3. Mini Project Code

```
board = [str(i) for i in range(1, 10)]

def print_board():
    print(board[0], "|", board[1], "|", board[2])
    print("--+---+--")
    print(board[3], "|", board[4], "|", board[5])
    print("--+---+--")
    print(board[6], "|", board[7], "|", board[8])

def check_winner():
    wins = [(0,1,2),(3,4,5),(6,7,8),
            (0,3,6),(1,4,7),(2,5,8),
            (0,4,8),(2,4,6)]
    for a, b, c in wins:
        if board[a] == board[b] == board[c]:
            return board[a]
```

```
    return None

current_player = "X"

while True:
    print_board()
    move = input("Player " + current_player + ", pick a position (1-9): ")

    if not move.isdigit() or int(move) not in range(1, 10):
        print("Invalid position. Try again.")
        continue

    index = int(move) - 1
    if board[index] in ("X", "O"):
        print("That spot is taken. Try again.")
        continue

    board[index] = current_player
    winner = check_winner()

    if winner:
        print_board()
        print("Player", winner, "wins!")
        break

    if all(cell in ("X", "O") for cell in board):
        print_board()
        print("It's a draw!")
        break

    current_player = "O" if current_player == "X" else "X"
```

4. Sample Execution

```
1 | 2 | 3
--+---+--
4 | 5 | 6
--+---+--
7 | 8 | 9
Player X, pick a position (1-9): 5
X | 2 | 3
--+---+--
4 | X | 6
--+---+--
7 | 8 | 9
Player O, pick a position (1-9): 1
...
Player X wins!
```

5. Result

The two-player Tic-Tac-Toe game was successfully implemented and tested. The program correctly displayed the board after every move, validated player input, alternated turns between X and O, detected all winning combinations, identified a draw when the board filled up, and announced the final result.